NETWORKING LESSON REVIEWER

Computer Networking Fundamentals

This reviewer introduces the fundamental concepts of computer networking, including network types, topologies, devices, transmission media, IP addressing, network models, protocols, security, and troubleshooting.

1. Introduction to Computer Networking

A computer network is a group of interconnected computers and devices that communicate and share resources such as files, printers, applications, and Internet access. Networking allows users and systems to exchange data efficiently.

- Node – any device connected to a network.
- Client – a device or program that requests a service.
- Server – a computer or system that provides services or resources.
- Network – interconnected devices that communicate using agreed rules called protocols.

2. Types of Networks

Networks can be classified according to their size, coverage, and purpose.

- PAN (Personal Area Network) – a small network around one person, such as a phone connected to a smartwatch.
- LAN (Local Area Network) – covers a limited area such as a classroom, office, or home.
- MAN (Metropolitan Area Network) – connects networks across a city or metropolitan area.
- WAN (Wide Area Network) – covers large geographic areas; the Internet is the most well-known example.

3. Network Topologies

A network topology describes how devices are arranged and connected.

- Bus – devices share a common backbone cable.
- Star – devices connect to a central switch or hub.
- Ring – devices form a circular connection.
- Mesh – devices have multiple interconnections, providing redundancy.
- Tree – devices are arranged in a hierarchical structure.
- Hybrid – combines two or more topology types.

4. Network Devices

Network devices connect computers and control or direct network traffic.

- Router – connects different networks and forwards packets between them.
- Switch – connects devices within a LAN and forwards frames to the appropriate device.
- Hub – broadcasts incoming data to connected ports; largely replaced by switches.
- Modem – converts signals so a network can communicate with an Internet service.

- Access Point (AP) – provides wireless network access to devices.
- Repeater – regenerates or retransmits a signal to extend its range.
- Firewall – filters network traffic according to security rules.

5. Transmission Media

Transmission media are the paths used to carry data between network devices.

- Twisted-pair cable – common Ethernet cable such as UTP.
- Coaxial cable – uses a central conductor and shielding; used in some broadband and legacy networks.
- Fiber-optic cable – transmits data using light; supports high speeds and long distances.
- Wireless – uses radio waves, such as Wi-Fi and Bluetooth, instead of physical cables.

6. IP Addressing

An IP address identifies a device or network interface for communication on an IP network.

- IPv4 – uses 32-bit addresses, commonly written in dotted-decimal form, such as 192.168.1.10.
- IPv6 – uses 128-bit addresses and was designed to provide a much larger address space.
- Private IPv4 addresses are commonly used inside local networks, while public addresses can be routed on the Internet.
- Subnet mask/prefix – indicates which part of an IP address represents the network and which part identifies hosts.
- Default gateway – the device, usually a router, used to reach other networks.

7. MAC Address

A MAC (Media Access Control) address is a link-layer identifier associated with a network interface. It is commonly represented as hexadecimal pairs, for example 00:1A:2B:3C:4D:5E. MAC addresses are used for communication on local network segments.

8. OSI Model

The OSI model divides network communication into seven conceptual layers.

- 7. Application – network services used by applications.
- 6. Presentation – data representation, encoding, compression, and encryption concepts.
- 5. Session – establishes and manages communication sessions.
- 4. Transport – end-to-end delivery, reliability, and segmentation; examples include TCP and UDP.
- 3. Network – logical addressing and routing; IP operates here.
- 2. Data Link – frames, MAC addressing, and local delivery.
- 1. Physical – electrical, optical, or radio signals and physical media.

9. TCP/IP Model

The TCP/IP model is the practical networking model used by the Internet.

- Application – includes protocols such as HTTP, HTTPS, DNS, SMTP, and FTP.
- Transport – TCP and UDP provide transport services.

- Internet – IP provides addressing and routing.
- Network Access/Link – handles local network communication and physical transmission.

10. Common Network Protocols

Protocols define rules for how devices communicate.

- HTTP – transfers web content.
- HTTPS – HTTP protected with encryption using TLS.
- DNS – translates domain names into IP addresses.
- DHCP – automatically provides network configuration such as IP addresses.
- TCP – connection-oriented transport with reliability and ordered delivery.
- UDP – connectionless transport with lower overhead and no built-in delivery guarantee.
- FTP – transfers files; secure alternatives such as SFTP are commonly preferred for protected transfers.
- SMTP – sends email between mail systems.

11. Network Security

Network security protects systems, communications, and data from unauthorized access, misuse, and disruption.

- Use strong, unique passwords and multi-factor authentication where available.
- Keep operating systems, applications, routers, and security software updated.
- Use firewalls and properly configured access controls.
- Use encryption such as HTTPS and secure Wi-Fi configurations.
- Avoid opening suspicious links or attachments.
- Back up important data and protect backups from unauthorized access.

12. Internet and Intranet

The Internet is a global interconnected network of networks. An intranet is a private network used within an organization. An extranet may provide controlled access to selected external users or partners.

13. Basic Network Troubleshooting

Network troubleshooting is the process of identifying and resolving connectivity or performance problems.

- Check physical cables, power, and link/activity lights.
- Confirm Wi-Fi is enabled and the correct network is selected.
- Check IP configuration using tools such as ipconfig on Windows.
- Test local connectivity with ping.
- Check whether the default gateway is reachable.
- Test DNS resolution when websites cannot be reached by name.
- Restart or reconnect affected equipment when appropriate.
- Check firewall, router, access-point, and service status before changing advanced settings.

Quick Review Questions

1. What is a computer network?

2. What is the difference between a LAN and a WAN?

3. What device connects different networks and forwards packets?

4. What is the purpose of a switch?

5. What is the main difference between wired and wireless transmission?

6. How many bits are in an IPv4 address?

7. What is the purpose of a default gateway?

8. What does DNS do?

9. What is the role of DHCP?

10. Which OSI layer is responsible for routing and logical addressing?

11. What is the difference between TCP and UDP?

12. What is the purpose of a firewall?

13. What is the difference between the Internet and an intranet?

14. What command can be used in Windows to view IP configuration?

15. What command can be used to test basic network reachability?

Answer Key

1. A network is a group of interconnected devices that communicate and share resources.

2. LAN covers a limited local area; WAN covers a much larger geographic area.

3. Router.

4. A switch connects devices in a LAN and forwards frames to the intended device.

5. Wired uses physical media; wireless uses radio communication.

6. 32 bits.

7. It provides a path from the local network to other networks.

8. DNS translates domain names into IP addresses.

9. DHCP automatically provides network configuration, including IP addresses.

10. Layer 3, the Network layer.

11. TCP provides connection-oriented, reliable, ordered delivery; UDP has lower overhead and does not provide the same built-in reliability.

12. It filters network traffic according to security rules.

13. The Internet is a global network of networks; an intranet is a private organizational network.

14. ipconfig.

15. ping.

Key Terms to Remember

Term	Meaning
LAN	Local Area Network
WAN	Wide Area Network
IP	Internet Protocol
MAC	Media Access Control
DNS	Domain Name System
DHCP	Dynamic Host Configuration Protocol
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
HTTP	Hypertext Transfer Protocol
HTTPS	HTTP over TLS
OSI	Open Systems Interconnection